Largen Animal Surgery and Practice

Presurgical consideration & anesthesia_2



Infiltration / Regional analgesia

Regional analgesia results form desensitization of sensory nerve to a given area. This may be performed by infiltration into the desired location of by blocking sensory nerve innervating a region. Depending on the required duration of anesthesia, local anesthetic agents including lidocaine (short onset and duration), mepivacaine (rapid onset with intermediate duration), bupivacaine (long onset and duration) may be used.

Infiltration / Regional analgesia

Many active ingredients (core structures) of anesthetics are organic weak bases, which are poorly water-soluble and unstable. Converting them into hydrochloride salts (HCl) improves water solubility, enhances absorption and bioavailability, and stabilizes the chemical structure (extending shelf life). However, because these aqueous solutions are slightly acidic, they can cause temporary pain and a burning sensation at the injection site.

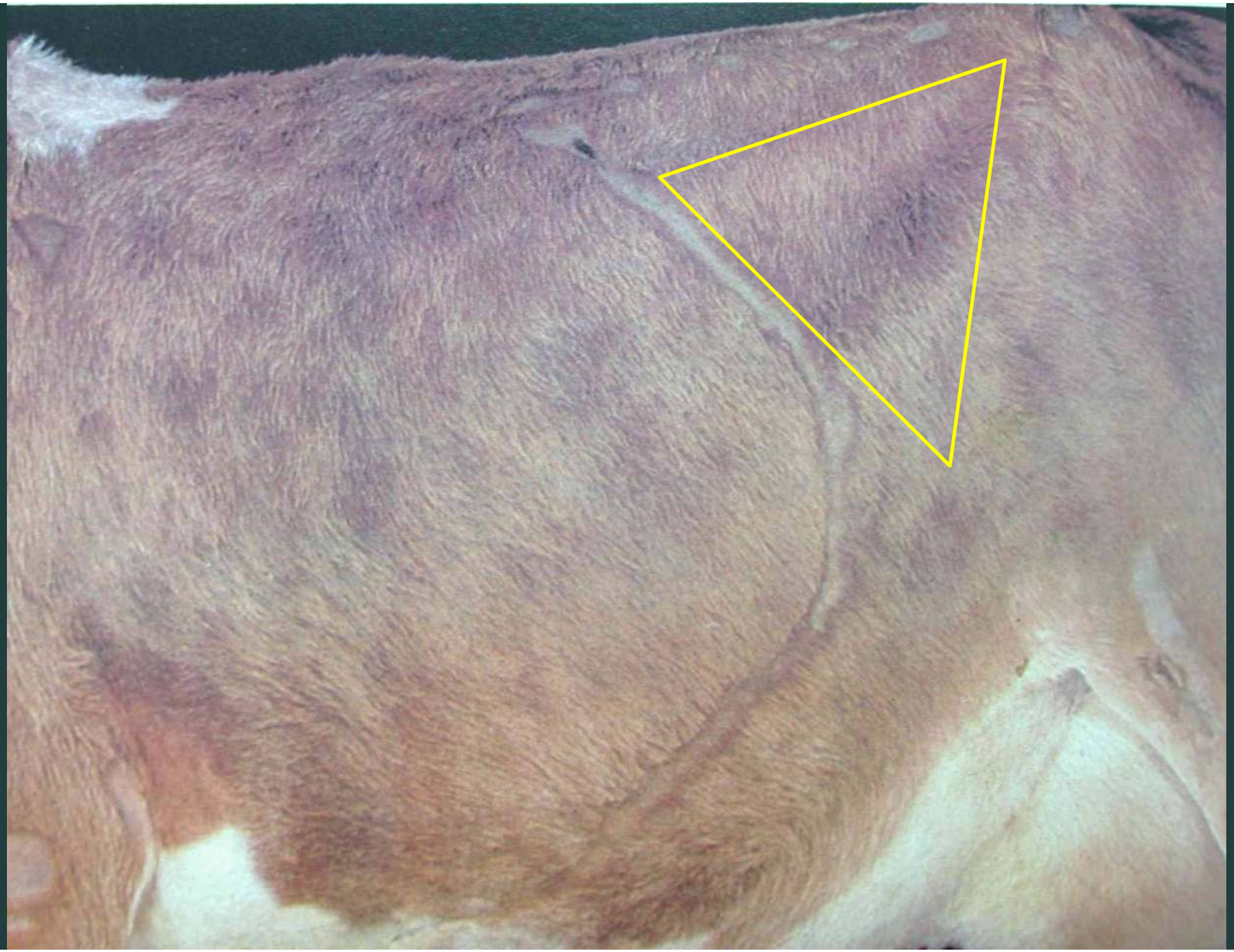
Infiltration / Regional analgesia

Due to cardiovascular toxicity with vascular absorption, bupivacaine use is usually limited to epidural and perineural administration. Lidocaine and mepivacaine may be used by any route.

Excessive concentrations of local anesthetics in the blood can cause systemic toxicity, including inhibiting sodium ion channels on brain and heart cell membranes, affecting CNS and CVS.

Use of Lidocaine

The safe total dose of lidocaine is usually 3–5 mg/kg (90–150 mL of 2% lidocaine/600 kg cattle). Do not exceed the dose at one time. Lidocaine takes effect about 5–10 minutes after injection, and the analgesic effect lasts for about 90–180 minutes. Clinically, about 100 mL of 2% lidocaine is sufficient for general surgery on an adult cow.



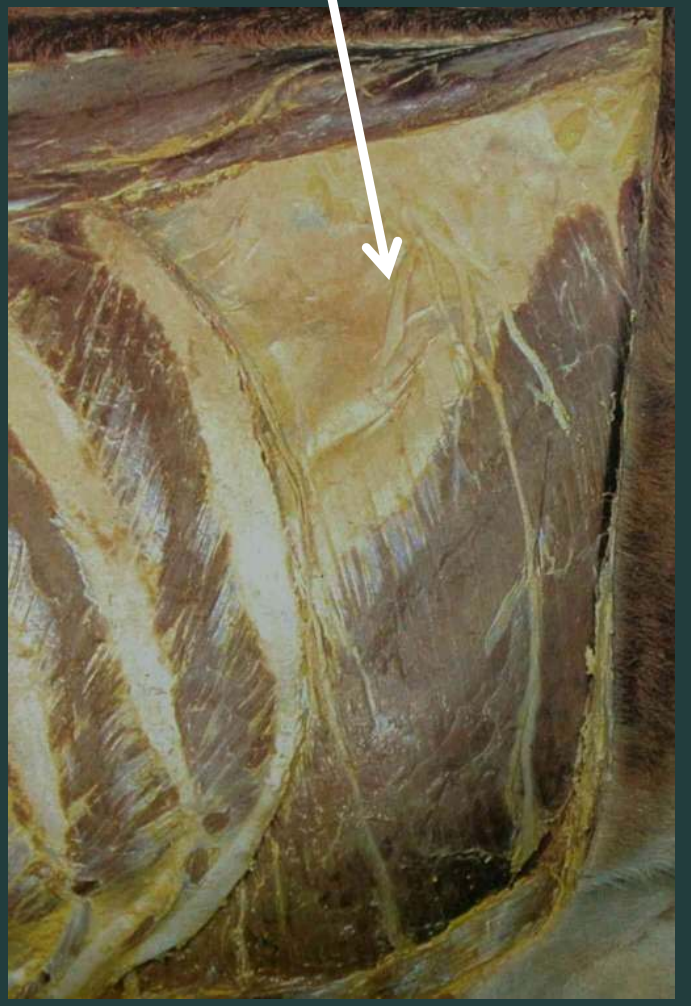
external oblique



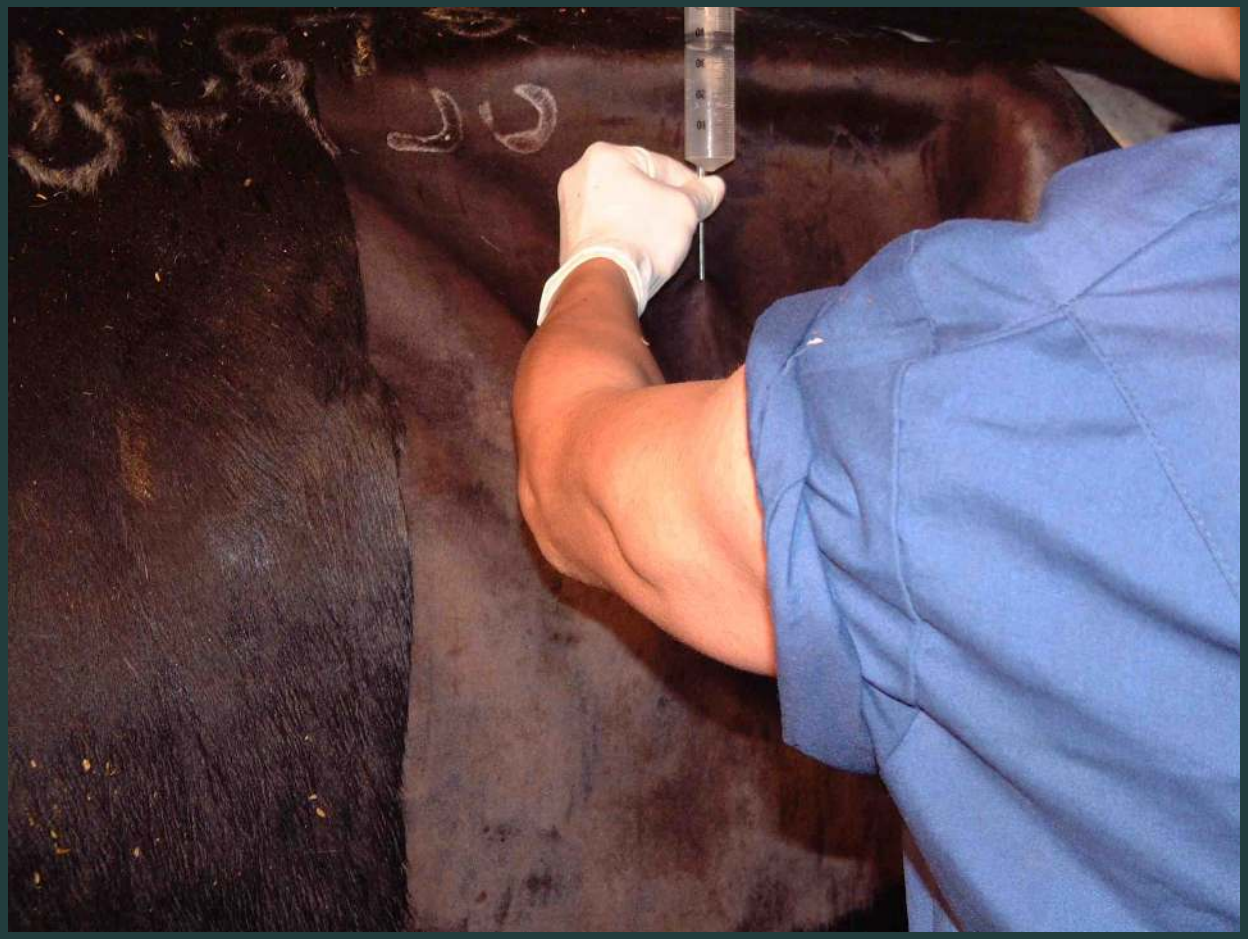
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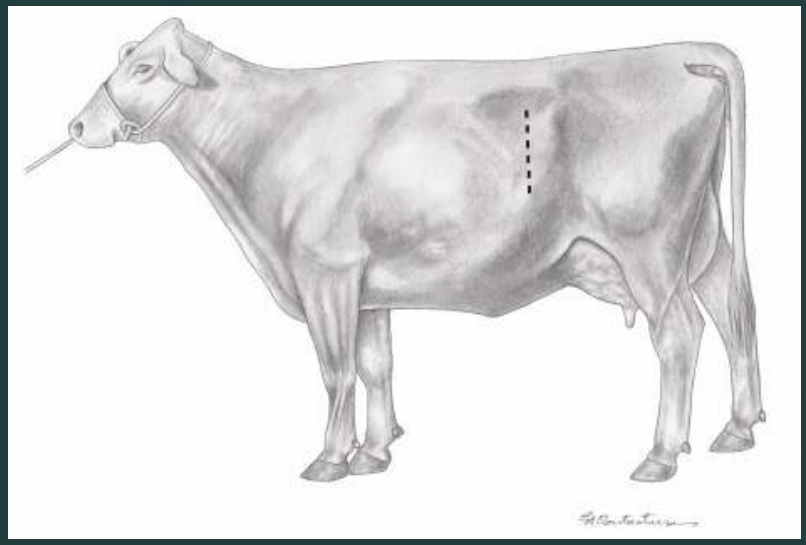


Transversus abdominis



Linear Infiltration Anesthesia

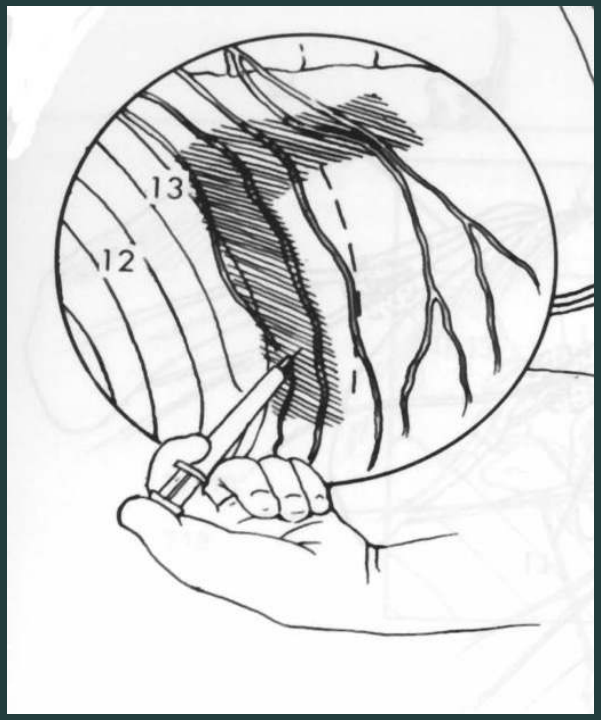






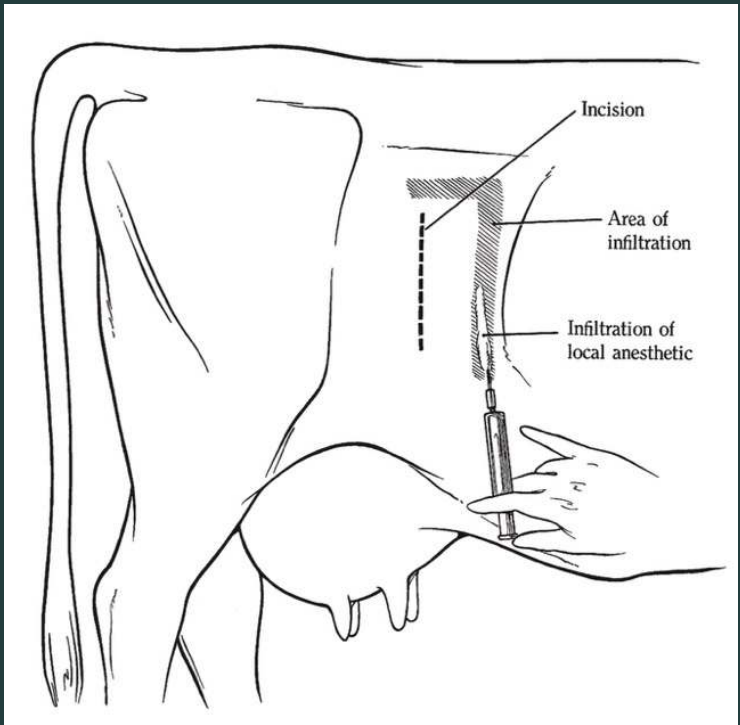
Infiltration needle
Dr. Wille's model, with side openings and small outlet,
140 mm long, 1,8 mm Ø,stainless

Inverted L Block



- Desensitizes the areas innervated by the 13th thoracic nerve (T13) and the 1st and 2nd lumbar nerves (L1 & L2).
- applicable for Caesarean section or rumen fistulation.

Inverted L Block



- Horizontal line approximately 5–10 cm ventral to the lumbar transverse processes
- Vertical line 5–10 cm posterior to the last rib
- Anesthetic blockade across one subcutaneous layer and two muscular layers and peritoneum.

Infiltration Anesthesia

- Significant amounts of local anesthetic inject into the peritoneal cavity should be avoided as rapid vascular absorption can result in toxicity.
- Injection should be made in straight lines.
- Infiltration of the anesthetic into the incision line causes edema in the tissues and may affect wound healing.

Linear/ Inverted L Block



- Inject approximately 3–5 mL of lidocaine at the uppermost point of the incision using an 18-G needle.
- Switch to a 15-G, 14-cm long needle. Slightly elevate the skin and advance the long needle vertically downward into the subcutis until the entire needle is submerged.

Linear/ Inverted L Block

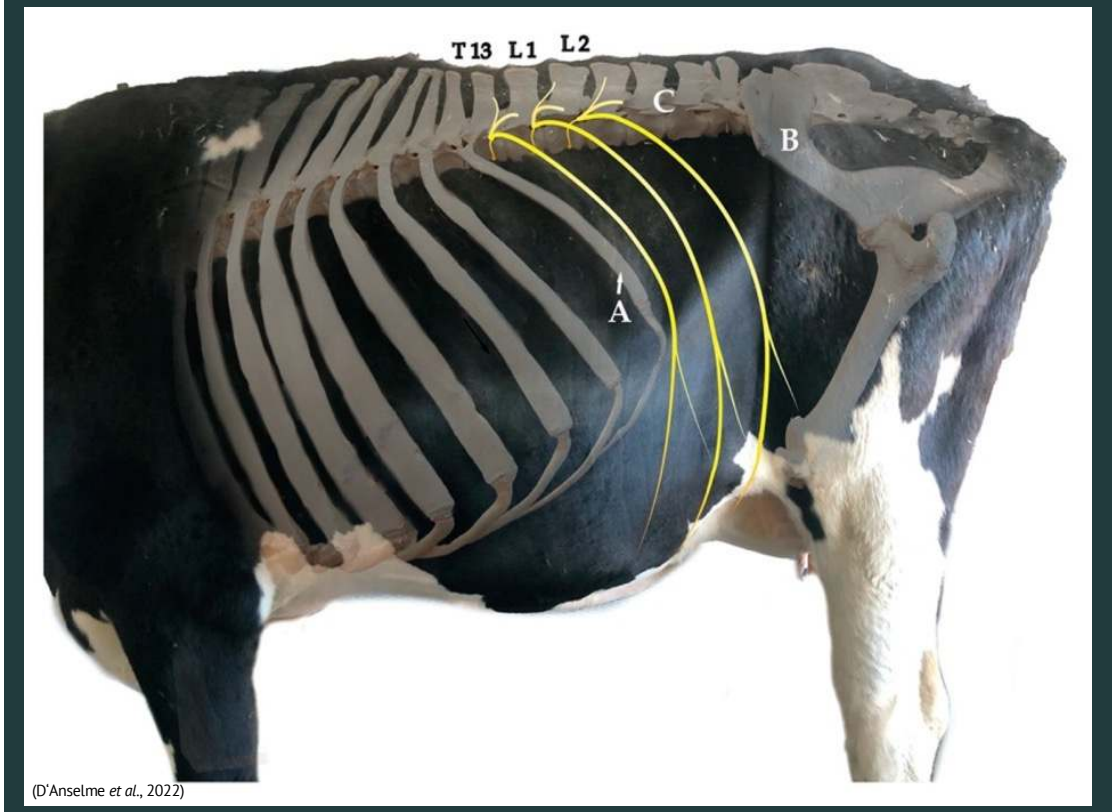


- Slowly inject the drug while concurrently retracting the needle to ensure a linear distribution along the incision site.
- Total volume of anesthetic used per layer should be about 25–30 mL.

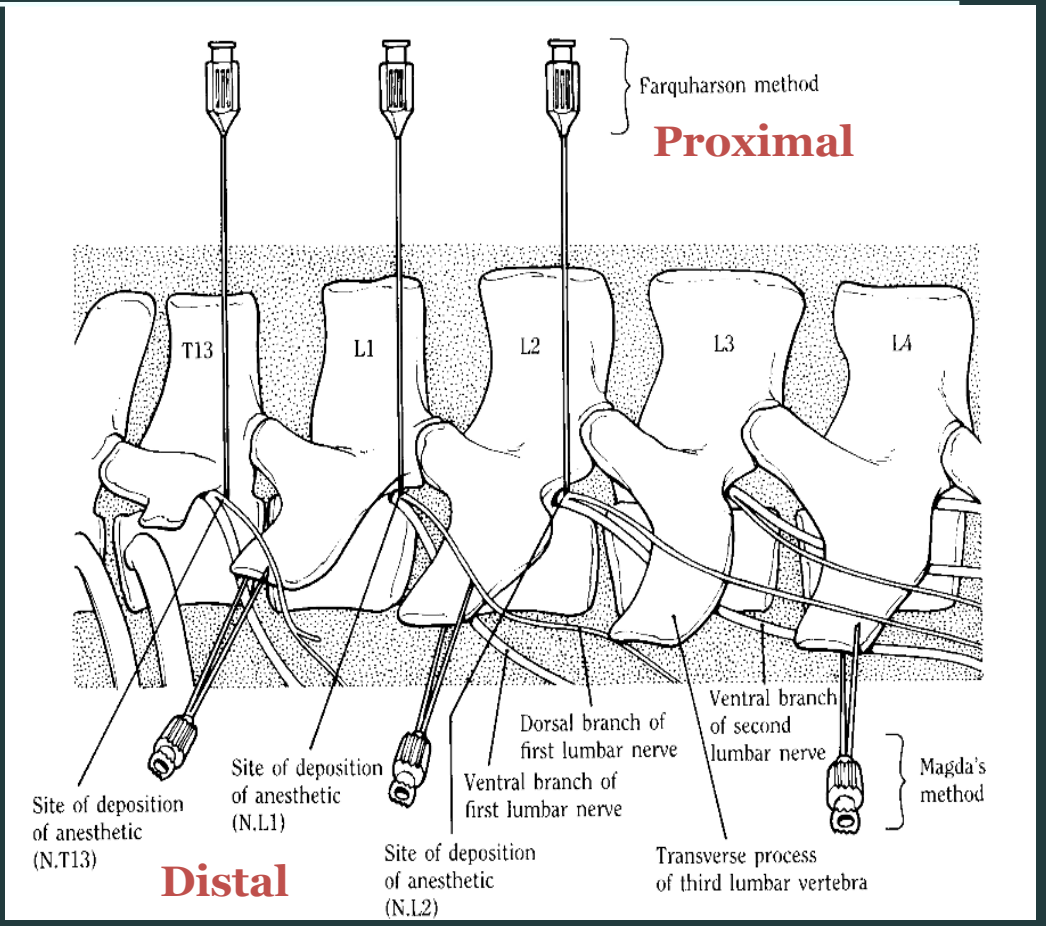
Linear/ Inverted L Block

- Upon completion of subcutaneous anesthesia, hold the long needle horizontally and insert it approximately 1–1.5 cm into the muscular layer.
- Advance the needle vertically downward through the muscle layer until the entire needle is submerged.
- Inject the anesthetic while concurrently withdrawing the needle, using the same technique described above.

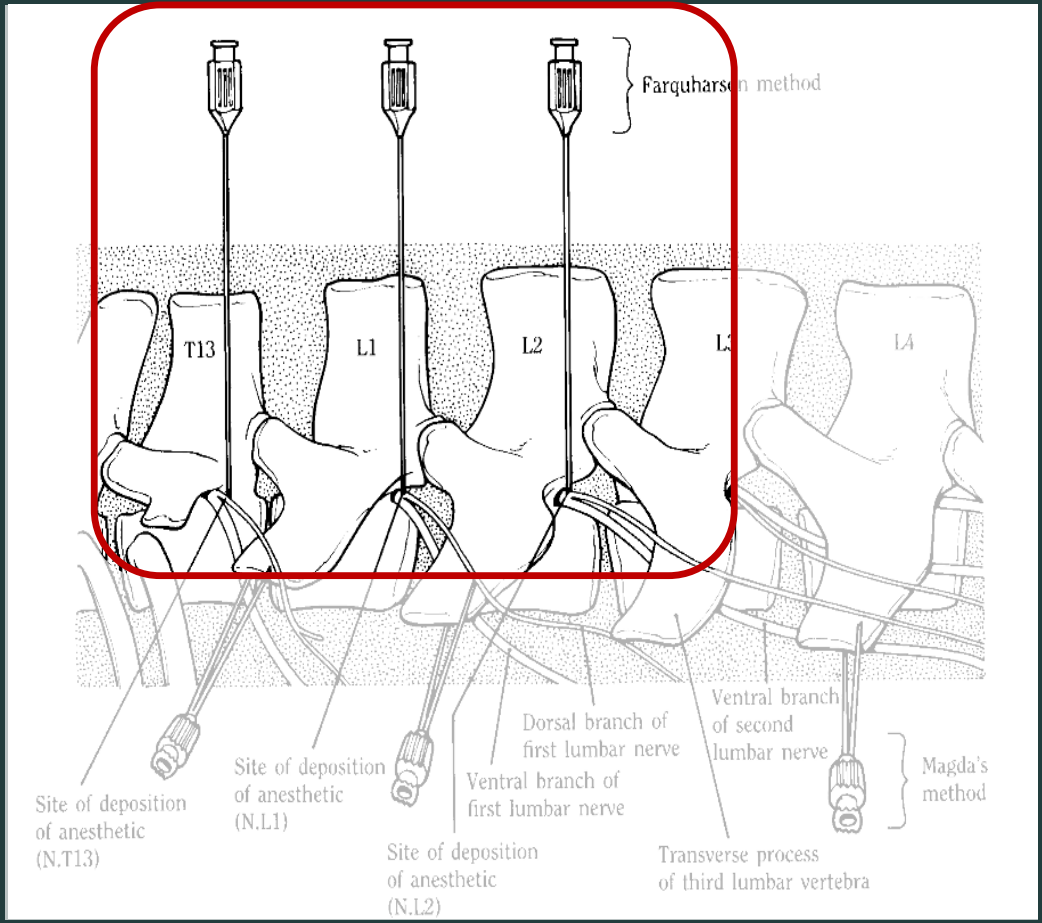
Paravertebral Block (脊椎旁神經阻斷法)



(D'Anselme *et al.*, 2022)

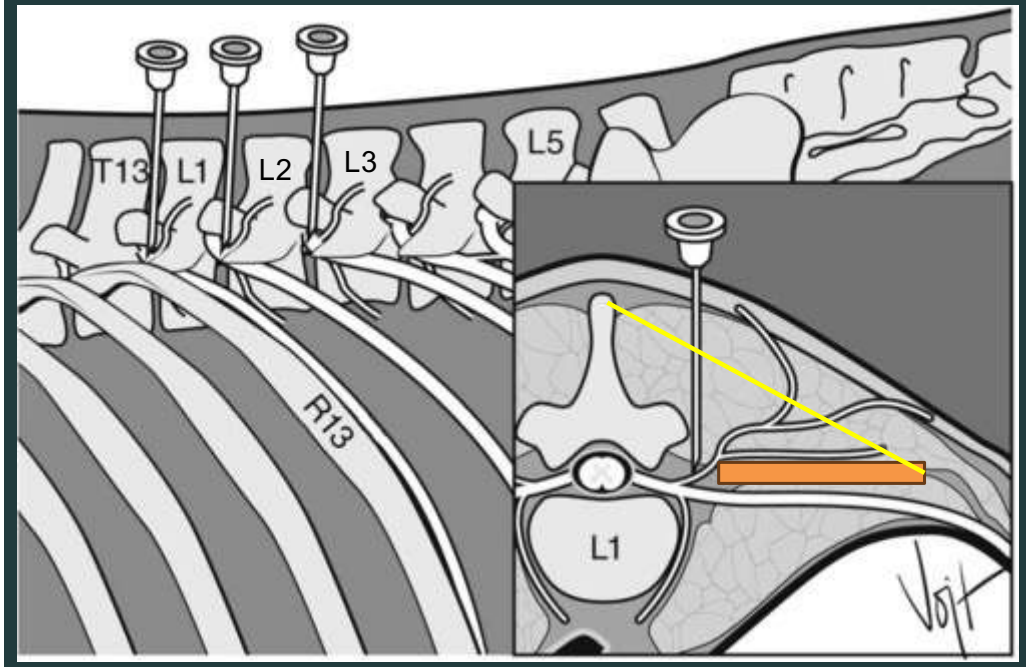


Proximal Paravertebral Block (近端脊椎旁神經阻斷)



- Target: T13, L1 and L2
- Intervertebral foramen (3–5 cm from the midline)
- Insertion caudal to the transverse processes of T13, L1, and L2

Proximal Paravertebral Block (近端脊椎旁神經阻斷)



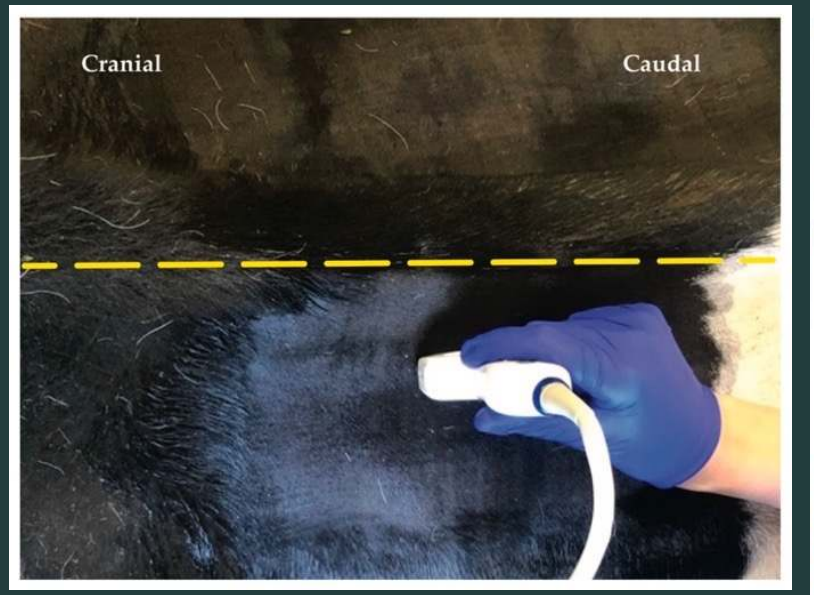
https://veteriankey.com/local-anesthesia-in-ruminants-and-pigs/

- Landmark:
  L1 transverse process → T13
  L2 → L1
  L3 → L2
- Inject site: 3–5 cm from the midline of the transverse process cranial edge to the dorsal midline.

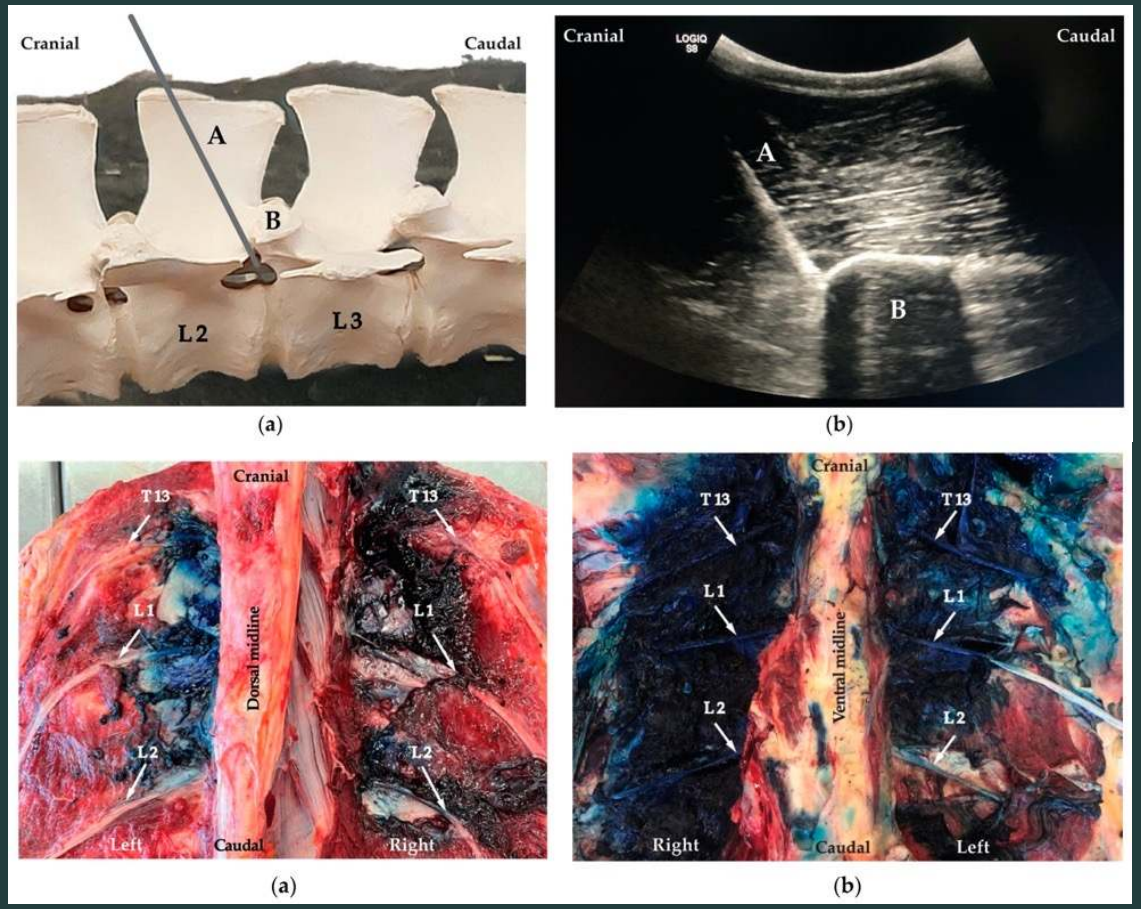
Proximal Paravertebral Block (近端脊椎旁神經阻斷)

- Inject a small amount of 2% lidocaine into each injection site.
- Use a 1-inch, 16-G needle as a guide, then use a 10-cm, 20-G needle to puncture the intertransverse ligament perpendicularly through the transverse process, and inject 10–15 mL of anesthetic.
- Pull the needle upward about 0.75–1.0 cm, and inject about 10 mL of anesthetic above the ligament as well.

Proximal Paravertebral Block (近端脊椎旁神經阻斷)



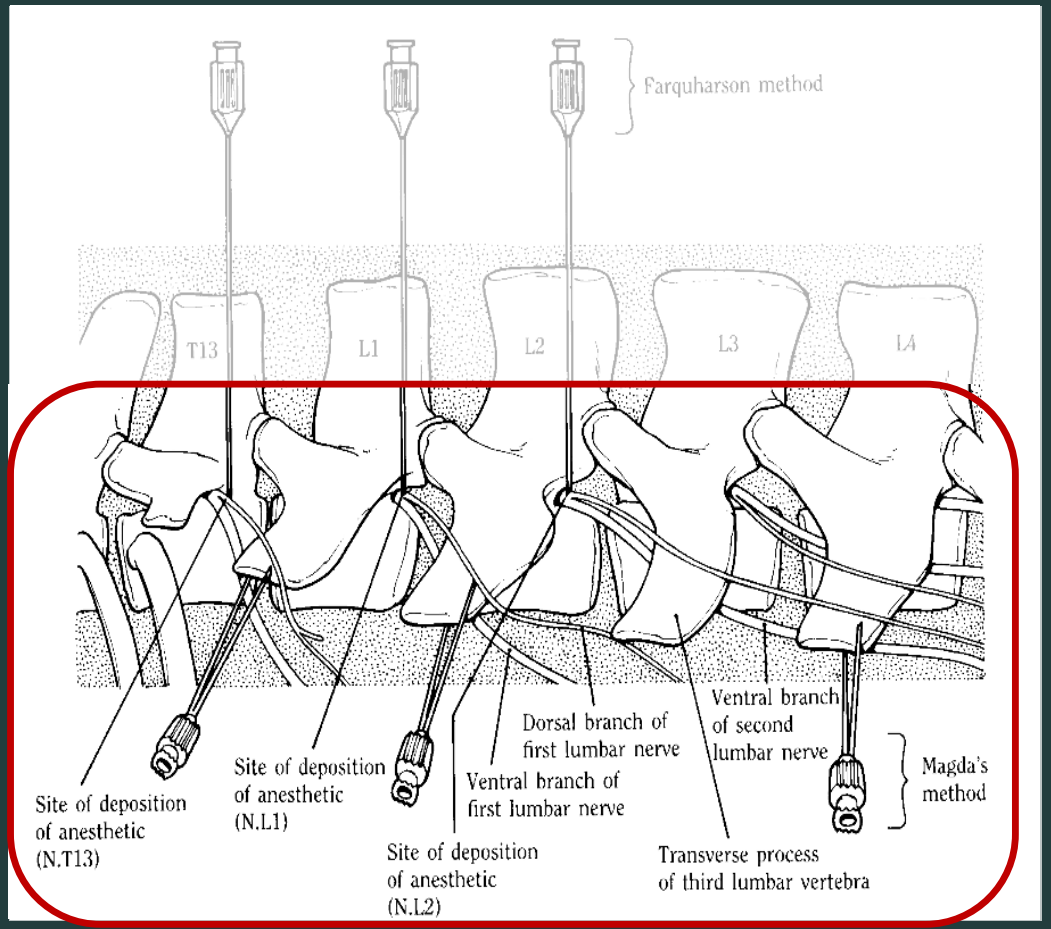
(D'Anselme *et al*., 2022)



Proximal Paravertebral Block (近端脊椎旁神經阻斷)

- Side effect:
  - A temporary lateral deviation of the spine due to muscle paralysis.
  - Vasodilation of surface vessels mat.

Distal Paravertebral Block (遠端脊椎旁神經阻斷)

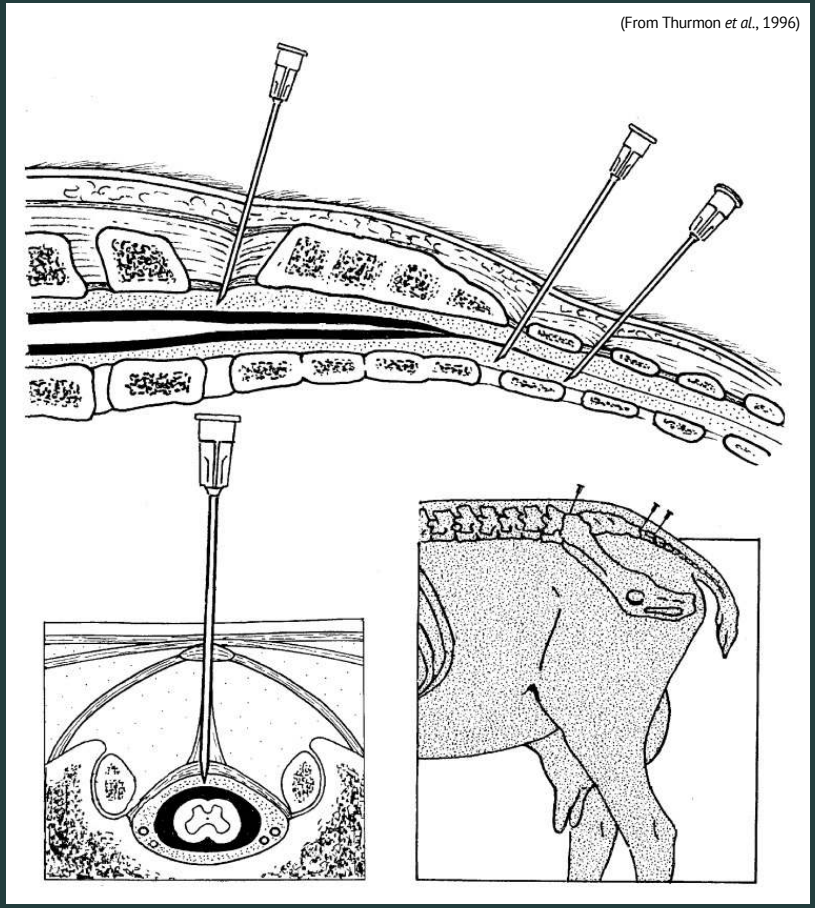


- Target: T13, L1, and L2
- Injection site: the ends of the L1, L2 and L4 transverse processes.
- Lateral deviation of the spine does not occur.

Distal Paravertebral Block (遠端脊椎旁神經阻斷)

- Insert an 18-G needle under transverse process towards the midline and cranial direction.
- Inject 10–15 mL of anesthetic (sector).
- Withdraw and redirect the needle slightly dorsal and caudal to the transverse process.
- Inject 10–15 mL of anesthetic (sector).

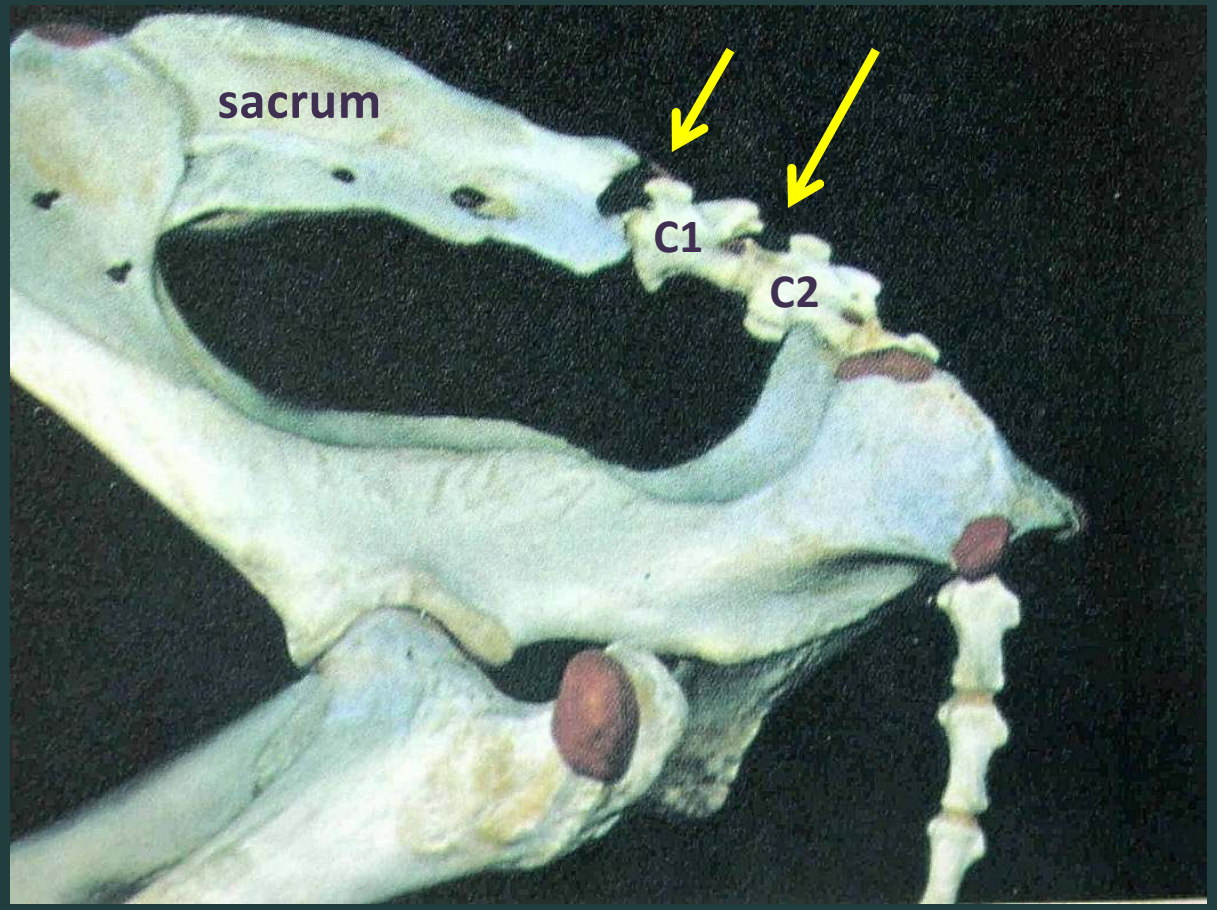
Epidural anesthesia (硬膜外麻醉)



- Lumbosacral (sheep, goats) and caudal (cattle, horses)
- For surgeries on the tail, anus, vulva, perineum, hind udder, scrotum, and upper hind limbs.
  - with/without concurrent sedation
- Use to treat and control tenesmus caused by enteritis or proctitis

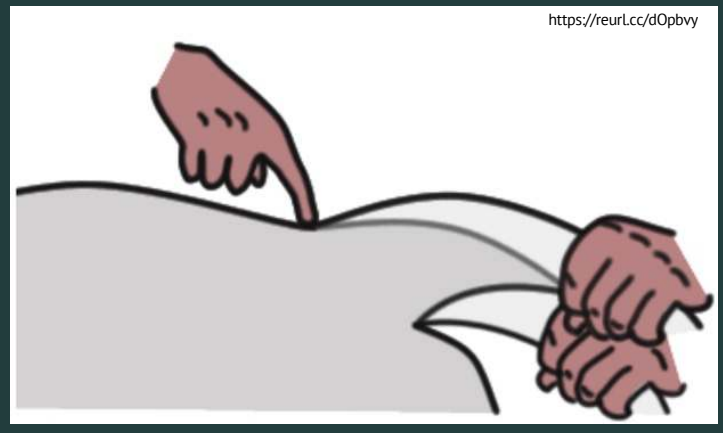
Caudal Epidural anesthesia (尾根硬膜外麻醉)

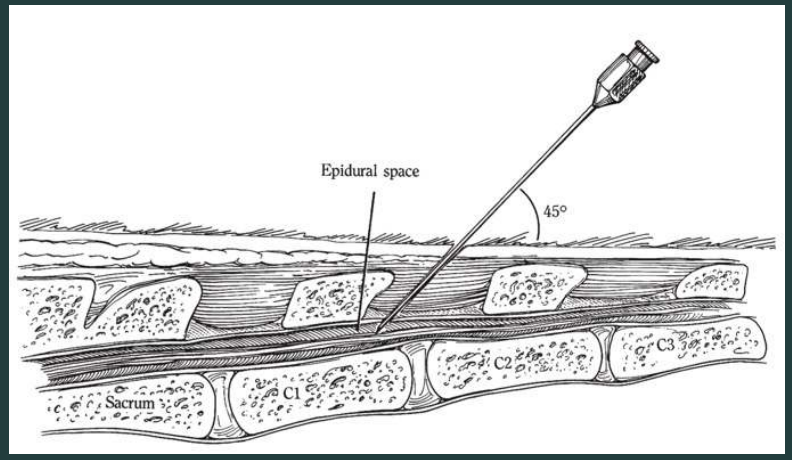
- Sacrococcygeal or 1st intercoccygeal space
- 2% lidocaine: 0.5–1.0 mL/50 kg BW (maximum: 10 mL/cattle; 3 mL/goats & sheep)



Caudal Epidural anesthesia (尾根硬膜外麻醉)

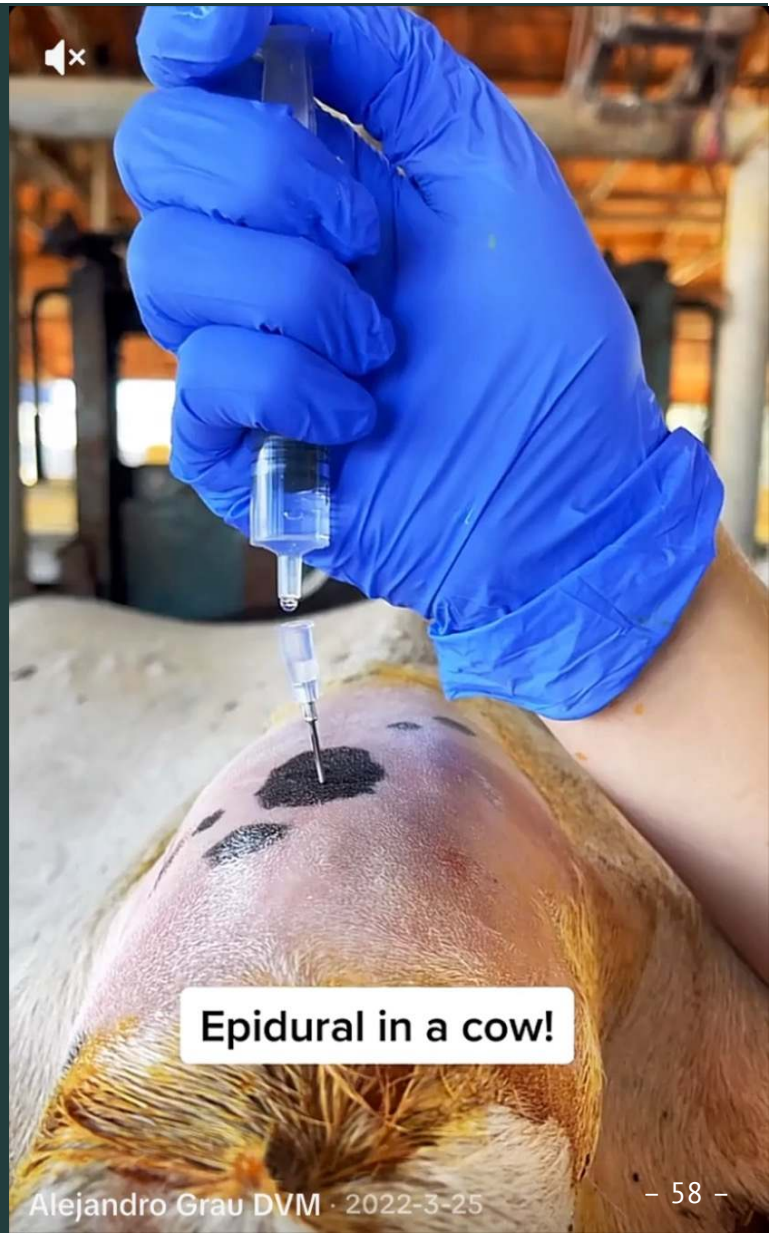
- Locate the space of C1 & C2: the first articulation caudal to the sacrum
- Clipping, skin preparation
- Introduce an 18- or 20-G needle
- through the center of the space on the midline at a 60–75° angle until its point hits the floor of the spinal canal.





Infiltration Anesthesia

- Drop a little anesthetic or saline into the needle hub.
- Retracted the needle slightly.
  - When the needle is inserted into the epidural space (especially the first time), the fluid will be drawn into the space.
- Inject the anesthetic without resistance

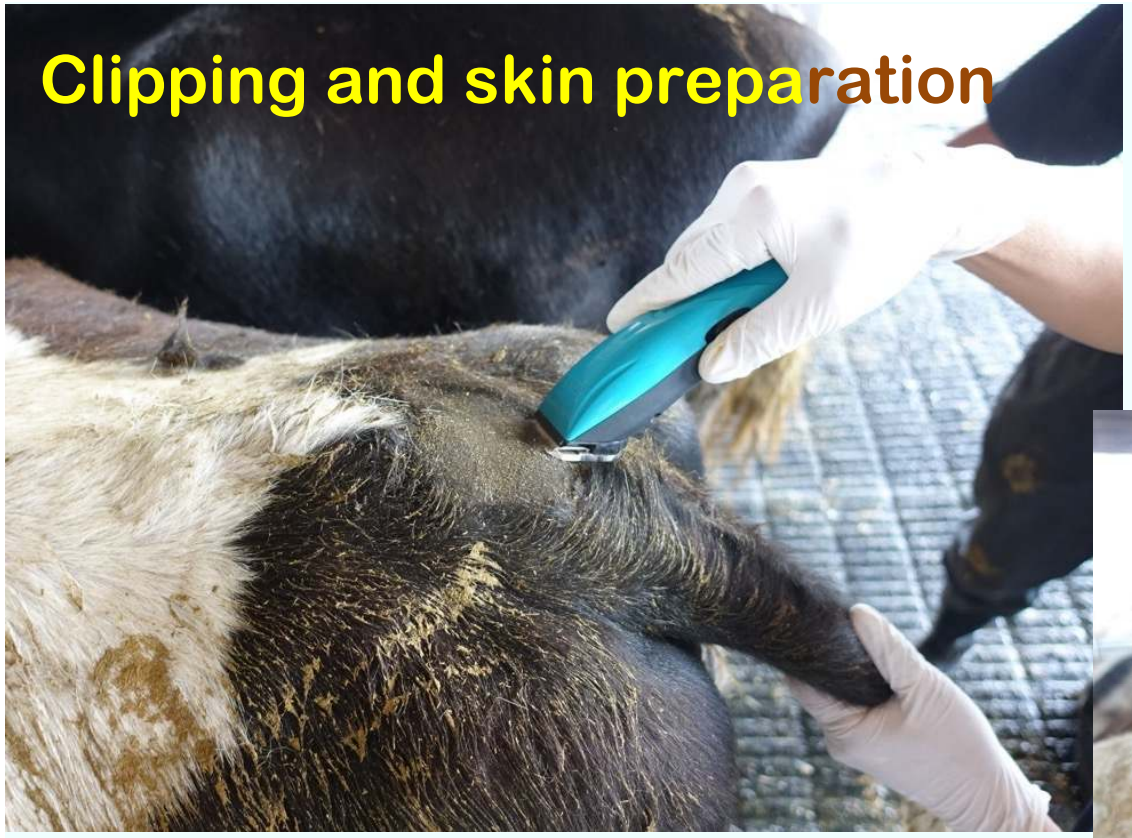




確認S5-Co1關節



Clipping and skin preparation



Insert an 18-G, 1¾” needle




hanging drop technique


Inject the agent